

(Material)Safety Data Sheet
According to Regulation (EC) No 1907/2006

Formic acid 85%

Version 1.2

Issue date: 05/02/2019

Revision date: 07/04/2019

(M)SDS Record Number: CSSS-TCO-010-113448

Section 1 Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier:

Identification on the label/Trade name: Formic acid 85%

Additional identification: CAS#64-18-6; EC#200-579-1

Identification of the product: See section 3

Index Number: 607-001-00-0

REACH registration No.: 01-2119491174-37-0010

1.2 Relevant identified uses of the substance and uses advised against:

1.2.1 Identified uses:

Manufacture of substance

Distribution of substance

Uses in Coatings

Use in Cleaning Agents

Use in Laboratories

Industrial manufacture of polymers, resins

Polymer processing

Use as processing aid

Use in Cleaning Agents

Use in Laboratories

Polymer processing

Use as processing aid

Animal nutrition

Use as preserving

Use in Cleaning Agents

Use as processing aid

1.2.2 Uses advised against:

Not available.

1.3 Details of the supplier of the safety data sheet:

Supplier(Only representative): Chemical Inspection and Regulation Service Limited

Supplier(Manufacturer): LIAOCHENG LUXI FORMIC ACID CHEMICAL CO., LTD

Address: Luxi Chemical Industry Park, Liaocheng Development Area, Shandong Province, China.

Contact person(E-mail): xiaozhengtao@luxichemical.de

Telephone: 0086 635 3487663

Fax: 0086 635 3481267

1.4 Emergency telephone Number:

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Available outside office hours?

YES

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NO

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Section 2 Hazards Identification

2.1 Classification of the substance/mixture:

2.1.1 Classification:

The 85% formic acid is classified as following according to 67/548/EEC and REGULATION (EC) No 1272/2008:

REGULATION (EC) No 1272/2008	
Hazard classes/Hazard categories	Hazard statement
Acute Tox-oral.4	H302
Skin Corr. 1B	H314
Acute Tox-inhalation.4	H332

For full text of H- phrases: see section 2.2.

67/548/EEC	
Hazards characteristics	R-Phrases
C	R35
Xn	R20/22

For full text of R- phrases: see section 16.

2.2 label elements:

Hazard Pictograms:



Signal Word(S):

Danger

Hazard Statement:

H302 Harmful if swallowed.

H314 Causes severe skin burns and eye damage.

H332 Toxic if inhaled

Precautionary statement:

P260 Do not breathe dust/fume/gas/mist/vapours/spray.

P264 Wash hands thoroughly after handling.

P270 Do not eat, drink or smoke when using this product.

P271 Use only outdoors or in a well-ventilated area.

P280 Wear protective gloves/protective clothing/eye protection/face protection.

P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.

P303 + P361 + P353 IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Rinse skin with water/shower.

P304 + P340: IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P310 Immediately call a POISON CENTER or doctor/physician.

P363 Wash contaminated clothing before reuse.

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.

P405 Store locked up.

P501 Dispose of contents/container to local regulation.

2.3 Other hazards:

EUH071: corrosive to the respiratory tract

Section 3 Composition/information on ingredients

Substance/Mixture:

Mixture

Product name: Formic acid 85%

Version #: 1.2

Revision date: 05-02-2015.

Issue date: 07-04-2015.

SDS EU

2 / 8

Ingredient(s):

Chemical Name	Index No.	Registration No.	CAS No.	EC No.	Concentration	Classification according to Regulation (EC) No. 1272/2008
Formic acid	N/A	01-2119491174-37-0010	64-18-6	200-579-1	85%	Flam.liq 3, H226; Acute tox.3, H331; Skin Corr. 1A, H314; Acute Tox. 4, H302
Water	N/A	N/A	7732-18-5	231-791-2	15%	Not classified

Section 4 First aid measures**4.1 Description of first aid measures:**

In all cases of doubt, or when symptoms persist, seek medical attention.

4.1.1 In case of inhalation:

Get medical aid immediately. Remove from exposure to fresh air immediately. If breathing is difficult, give oxygen. Do NOT use mouth-to-mouth resuscitation. If breathing has ceased apply artificial respiration using oxygen and a suitable mechanical device such as a bag and a mask.

4.1.2 In case of skin contact:

Get medical aid immediately. Immediately flush skin with plenty of soap and water for at least 15 minutes while removing contaminated clothing and shoes. Wash clothing before reuse. Destroy contaminated shoes.

4.1.3 In case of eyes contact:

Get medical aid immediately. Do NOT allow victim to rub or keep eyes closed. Extensive irrigation with water is required (at least 30 minutes).

4.1.4 In case of ingestion:

Do NOT induce vomiting. If victim is conscious and alert, give 2-4 cupfuls of milk or water. Never give anything by mouth to an unconscious person. Get medical aid immediately.

4.2 Most important symptoms and effects, both acute and delayed:

Harmful if swallowed. Causes severe skin burns and eye damage. Harmful if inhaled.

4.3 Indication of any immediate medical attention and special treatment needed:

If skin irritation or rash occurs, get medical advice/attention.

Section 5 Fire-Fighting measures**5.1 Extinguishing media:****Suitable extinguishing media:**

For small fires, use dry chemical, carbon dioxide, or water spray. For large fires, use dry chemical, carbon dioxide, alcohol-resistant foam, or water spray.

Unsuitable extinguishing media:

Do NOT get water inside containers.

5.2 Special hazards arising from the substance or mixture

During a fire, irritating and highly toxic gases may be generated by thermal decomposition or combustion. Vapors may be heavier than air. They can spread along the ground and collect in low or confined areas. May polymerize explosively when involved in a fire. Contact with metals may evolve flammable hydrogen gas.

5.3 Special fire fighting methods and special protective actions for fire-fighters:

As in any fire, wear a self-contained breathing apparatus in pressure-demand, MSHA/NIOSH (approved or equivalent), and full protective gear. Use water spray to keep fire-exposed containers cool.

Section 6 Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures:

6.1.1 For non-emergency personnel: Remove ignition sources. Provide adequate ventilation. Avoid inhalation of vapour or dust. Avoid skin and eye contact.

6.1.2 For emergency responders: Wear an appropriate NIOSH/MSHA approved respirator

6.2 Environmental Precautions:

Avoid disposing into drainage/sewer system or directly into the aquatic environment.

6.3 Methods for Containment and Cleaning up:

Large spills may be neutralized with dilute alkaline solutions of soda ash, or lime. Avoid runoff into storm sewers and ditches which lead to waterways. Clean up spills immediately, observing precautions in the Protective Equipment section. Remove all sources of ignition. Absorb spill using an absorbent, non-combustible material such as earth, sand, or vermiculite. Do not use combustible materials such as saw dust. Use a spark-proof tool. Provide ventilation. Do not get water inside containers.

6.4 Reference to other sections:

See Section 7 for information on safe handling.

See section 8 for information on personal protection equipment.

See Section 13 for information on disposal.

6.5 Additional information:

Wet clean or vacuum up solids.

Don't use a brush or compressed air for cleaning surfaces or clothing.

Clear spills immediately.

Section 7 Handling and storage

7.1 Precautions for safe handling:

7.1.1 Protective measures:

Wash thoroughly after handling. Remove contaminated clothing and wash before reuse. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous

7.1.2 Advice on general occupational hygiene:

Use only in a well-ventilated area. Contents may develop pressure upon prolonged storage. Keep container tightly closed. Do not get on skin or in eyes. Do not ingest or inhale. Discard contaminated shoes. Do not pressurize, cut, weld, braze, solder, drill, grind, or expose empty containers to heat, sparks or open flames

7.2 Conditions for safe storage, including any incompatibilities:

Store in a tightly closed container. Keep from contact with oxidizing materials. Store in a cool, dry, well-ventilated area away from incompatible substances. Corrosives area. Do not store in metal containers. Do not store near alkaline substances. Vent periodically.

7.3 Specific end use(s):

Not applicable.

Section 8 Exposure Controls/Personal Protection

8.1 Control parameters:

8.1.1 Occupational exposure limits: TWA = 5ppm 9.4mg/m³; STEL =10ppm 19mg/m³

8.1.2 Additional exposure limits under the conditions of use: Not available.

8.1.3 DNEL/DMEL and PNEC-Values: Not available.

8.2 Exposure controls:

8.2.1 Appropriate engineering controls:

A system of local and/or general exhaust is recommended to keep employee exposures as low as possible. Local exhaust ventilation is generally preferred because it can control the emissions of the contaminant at its source, preventing dispersion of it into the general work area. Use a flame proof exhaust ventilation system.

8.2.2 Individual protection measures, such as personal protective equipment:

Eyeface protection	Tightly fitting safety goggles and face shield EN166.
Hand protection	Chemical resistant gloves meeting EN374. Chloroprene and butyl rubber may be suitable glove material.
Body protection	Wear chemical resistant apron and boots EN465.
Respiratory protection:	In case of inadequate ventilation or risk of inhalation of dust, use suitable respiratory equipment. Recommended filter type: Filter P2.
Thermal hazards:	Wear suitable protective clothing to prevent heat.

8.2.3 Environmental exposure controls:	Avoid discharge into the environment. This material and its container must be disposed of as hazardous waste. According to local regulations, Federal and official regulations.
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Section 9 Physical and chemical properties

9.1 Information on basic physical and chemical properties:

Appearance:	Liquid
Colour:	Colorless
Odour:	pungent odor - benzaldehyde-like
Odour threshold:	Not available
pH:	Not available
Melting point/range (°C):	8°C
Boiling point/range (°C):	100.8°C
Flash point (°C):	42 °C(Formic acid CAS#64-18-6 99%)
Evaporation rate:	Not available
Flammability limit - lower (%):	Not available
Flammability (solid, gas):	Not available
Ignition temperature (°C):	Not available
Upper/lower flammability/explosive limits:	Not available
Vapour pressure (20°C):	44.8 mm Hg @ 20 °C
Vapour density:	1.59
Relative Density:	Not available
Bulk density (kg/m³):	Not available
Water solubility (g/l):	Miscible
n-Octanol/Water (log Po/w):	Not available
Auto-ignition temperature:	ca.400°C(CAS#57-11-4)
Decomposition temperature:	Not available
Viscosity, dynamic (mPa.s):	1.607mPa @ 25 °C
Explosive properties:	Lower: 18.0 vol % Upper: 57.0 vol %

9.2. Other information:

Fat solubility(solvent– oil to be specified)	Not available
etc:	
Surface tension:	71.5mN/m at 20 °C
Dissociation constant in water(pKa):	pKa = 3.70 at 20 °C
Oxidation-reduction Potential:	Not available

Specific gravity:

Not available

Section 10 Stability and reactivity

10.1 Reactivity:

The substance is stable under normal storage and handling conditions.

10.2 Chemical stability:

Under normal conditions, the product is stable. No hazardous reaction when handled and stored according to provisions.

Hazardous reactions are not known.

10.3 Possibility of hazardous reactions:

Under normal conditions, not hazardous reactions will occur.

10.4 Conditions to avoid:

Incompatible materials, metals, excess heat, combustible materials, oxidizers, plastics.

10.5 Incompatible materials:

Strong oxidizing agents, strong bases, finely powdered metals, permanganates, sulfuric acid, hydrogen peroxides, nitromethane, furfuryl alcohol, hydrated thallium nitrate.

10.6 Hazardous decomposition products:

Carbon monoxide, irritating and toxic fumes and gases, carbon dioxide.

Section 11 Toxicological information

11.1 Toxicokinetics, metabolism and distribution:

Non-human toxicological data:

Not available

11.2 Information on toxicological effects:

Acute toxicity:

LD50(Oral, Rat):

730mg/kg

LD50(Dermal, Rabbit):

Not available

LC50(Inhalation, Rat):

74000mg/m³ air

Skin corrosion/Irritation:

Causes severe skin burns and eye damage.

Serious eye damage/irritation:

Not classified

Respiratory or skin sensitization:

Not classified

Germ cell mutagenicity:

Not classified

Carcinogenicity:

Not classified

Reproductive toxicity:

Not classified

STOT- single exposure:

Not classified

STOT-repeated exposure:

Not classified

Aspiration hazard:

Not classified

Section 12 Ecological information

Toxicity:

Acute toxicity		Time	Species	Method	Evaluation	Remarks
LC50	130mg/L	96h	Fish	OECD 203	N/A	N/A
EC50	365 mg/L	48h	Daphnia	OECD 202	N/A	N/A
EC50	1000mg/L	72h	Algae	OECD 201	N/A	N/A

In natural water it has been shown to adsorb to sediment and would probably also biodegrade.

Persistence and degradability:

Bioaccumulative potential:

Not bioaccumulative.

Mobility in soil:

Not available.

Results of PBT&vPvB assessment:

Not PBT or vPvB substance.

Other adverse effects:

Not applicable.

Section 13 Disposal considerations

13.1 Waste treatment methods:

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. US EPA guidelines for the classification determination are listed in 40 CFR Parts 261.3. Additionally, waste generators must consult state and local hazardous waste regulations to ensure complete and accurate classification. Federal, local or national regulations for proper disposal.

13.2 Product / Packaging disposal:

Disposal must be made according to official regulations.

Section 14 Transport information

	Land transport(ADR/RID)	Sea transport (IMDG)	Air transport (ICAO/IATA)
UN-Number	1779	1779	1779
UN Proper shipping name	FORMIC ACID	FORMIC ACID	FORMIC ACID
Transport hazard Class	8+3	8+3	8+3
Packaging group	II	II	II
Environmental hazards	FORMIC ACID	FORMIC ACID	FORMIC ACID
Special precautions for user	See section 2.2	See section 2.2	See section 2.2
Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code	IBC 02	IBC 02	IBC 02

Section 15 Regulation information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture:

Relevant information regarding authorization:

Not applicable.

Relevant information regarding restriction:

Not applicable.

Other EU regulations:

Employment restrictions concerning young person must be observed. For use only by technically qualified individuals.

Other National regulations:

Not applicable

Chemical Safety Assessment has been carried out?

YES

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NO

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Section 16 Other information

16.1 Indication of changes:

Version 1.2 Amended by EU No 453/2010

16.2 Relevant R- phrases (number and full text):

R35 Causes severe burns.

R20/22 Harmful by inhalation and if swallowed.

16.3 Training instructions:

Not applicable.

16.4 Further information:

This information is based upon the present state of our knowledge. This (M)SDS has been compiled and is solely intended for this

product.

16.5 Notice to reader:

Employers should use this information only as a supplement to other information gathered by them, and should make independent judgment of suitability of this information to ensure proper use and protect the health and safety of employees. This information is furnished without warranty, and any use of the product not in conformance with this (Material) Safety Data Sheet, or in combination with any other product or process, is the responsibility of the user.

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